

SurviveCovid-19++ : A collaborative healthcare game towards educating people about safety measures for Covid-19

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ABSTRACT

The potential of collaborative games, especially in the healthcare domain could be leveraged in educating people about the *Covid-19* pandemic that requires collaborative efforts from people with varied professional backgrounds. We present *SurviveCovid-19++*, a collaborative multiplayer desktop based game, to educate people about responsibilities of citizens with varied professional backgrounds, and emphasize on the need for collaboration to fight against the pandemic, by following safety measures. The game revolves around four roles delivering their duties, by following safety measures and collaboratively clearing multiple stages in the game. We evaluated the game through a qualitative and quantitative user survey, that presented encouraging results, with volunteers expressing increased empathy towards efforts of individuals with varied professional backgrounds, and better understanding of the importance of safety measures against *Covid-19*. We see that this work could provide insights to the CSCW audience on using collaborative games towards bringing awareness in the context of pandemics.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing systems and tools**; **Interactive systems and tools**.

KEYWORDS

Covid-19, Collaborative Game, Education, Safety Measures, Empathy

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1 INTRODUCTION

Games support effective learning and contribute to improved retention capacity [8, 12]. Games that involve collaborative aspects contribute to increased user engagement and consequently support better and motivated learning [3], and can be used in various contexts, including disaster response [2]. Collaborative aspects also improve communication in a team and thus support in enhancing the team spirit, while learning new concepts in the domain of interest of the game [4]. Such games have been introduced in various fields such as science and medicine and were observed to enhance the learning experience of users [13, 20].

With around eight strains of the *Covid-19* virus identified across the world¹, scientists are working towards developing vaccines to control each of these strains [9, 16]. The sanitation staff in majority of the countries are carrying out sanitization works in public places to create hygienic conditions, which could reduce the spread of the virus [6, 15]. Doctors and nurses across the world have been performing their duties impeccably, almost round the clock, to cure the patients effected by Covid-19 [17, 21]. With safety measures such as usage of masks and limited strength in public gatherings made mandatory in many countries, common people in across the world are also expected to follow the safety precautions and fulfill their duty of controlling the pandemic. Considering these varied duties and responsibilities of people with varied professional backgrounds, collaborative efforts and mutual support among all professions is essential to control the spread of the virus [5, 23]. This also emphasizes the need for individuals to understand the importance of each profession to control the pandemic and empathize with each other.

There exist very few games to bring awareness about Covid-19 and the safety measures to be followed against Covid-19. *SurviveCovid-19* [22] has been developed to bring awareness on the health measures to be followed to control Covid-19. However, it only supports single player version, and is only from a citizen's perspective. Considering the need for collaborative efforts to control the pandemic, and the advantages of collaborative games, a game that supports collaborative, multi-player version could better contribute in educating the users.

Hence, we present *SurviveCovid-19++*², a multi-player, collaborative game, to help in understanding collaborative effort required

¹<https://www.cdc.gov/coronavirus/2019-ncov/cases-updates/variant-surveillance/variant-info.html#Consequence>

²<https://vdheerajofficial.itch.io/survivecovid-19-plus-desktop>

by people with different professional backgrounds while including the aspects of safety measures, administering vaccines and multiple strains of the virus. We perform a preliminary evaluation on the influence of *SurviveCovid-19++* on players' understanding, player experience and usability. The results of the evaluation and the design of the game could provide insights to the CSCW audience on using collaborative games in the context of bringing awareness in the context of pandemics.

2 RELATED WORK

Collaborative and competitive games were observed to increase motivation among individuals to play educational games, which further improves their understanding on the topics being taught, and hence have also been adopted in the healthcare domain [1, 11, 14]. *StarsRace* has been developed as a collaborative mobile game with an aim to control obesity by requiring players to virtually collect stars placed at specific geographical locations, by moving around the map physically [1]. *LIVE* has been developed as a virtual environment platform to facilitate learning of self-management skills and provides social support to diabetic patients [14]. It presents a 3D view of various objects in the virtual environment that includes pharmacy items, fitness equipment, avatar choices and so on, while encouraging collaborative games and social networking throughout [14].

Some games, though limited in number, have been developed to bring awareness among individuals about pandemics. *Pandemic* game has been integrated with artificial intelligence to support players to travel across different places in the world in-game and collaboratively cure cities that get infected by various diseases [19]. *SurviveCovid-19* has been introduced to bring awareness among individuals on the safety measures to be followed to reduce the spread of *Covid-19* [22]. *COVID-19 - Did You Know?* has been developed as a true-false based question-answering themed game, to educate users about *Covid-19*, by providing information about various topics related to *Covid-19*, such as use of masks, social distancing and so on [10].

Of the limited number of games that consider pandemics, games that support collaborative game play are further less in number, which almost do not exist in the literature with respect to *Covid-19*. To the best of our knowledge, *Pandemic* [19] is the only collaborative game that considers pandemic aspect, but is not specific to *Covid-19*, and also does not include disease-specific elements in-game. Though controlling *Covid-19* requires collaborative efforts from individuals with varied professional backgrounds, we could not find any collaborative-games in the literature which attempt to educate about *Covid-19*, motivating us towards the *SurviveCovid-19++* game proposed here.

3 DESIGN OF SURVIVECOVID-19++

SurviveCovid-19++ is a multiplayer collaborative game developed based on the guidelines introduced for development of collaborative games in healthcare domain [4]. It can be played by a maximum of four players and a minimum of one player. It starts with the storyline of a group of friends in four different professions - *Doctor*, *Sanitation worker*, *Common man* (Citizen) and *Law Enforcer*, in a city, working towards controlling the spread of *Covid-19* and treating the

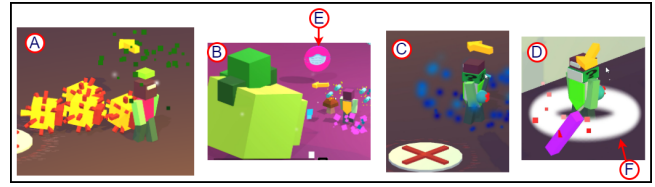


Figure 1: Snapshots of the game representing [A] Virus attacking citizen, [B] Stronger strain of the virus attempting to attack Doctor, [C] Sanitation worker leaving from hospital after being cured, [D] Sanitation worker collecting vaccine parts, [E] Masks to be collected by players when necessary, and [F] Sanitation worker guarded by a ring on due collection of mask.

affected individuals. It has defined roles (in-game profession-based) for each avatar and the players can select the avatar of their choice, without repetition in the team. Each avatar is assigned specific tasks, along with collaborative tasks that are to be accomplished to clear each stage of the game. The *Citizen* is required to collect groceries throughout the game. The *Doctor* is required to collect medicine and treat *Covid-19* infected patients with these medicines collected. The *Sanitation worker* needs to collect sanitizer/disinfectant refills throughout the game and use them to disinfect infected locations. The *Law Enforcer* is required to break crowds in the city. All the players are required to collect required number of vaccines apart from their personal goals in the game, while using masks and sanitizers to protect themselves from getting infected by the virus, in the game, to clear each stage. If an avatar gets infected by the virus, the health of the avatar gets reduced and the player's movement in the game is slowed down. The more the player gets infected, slower the movement of the player.

4 USER SCENARIO

Consider *Moksha* and her three friends interested in understanding collaborative efforts towards fighting *Covid-19*. They consider playing *SurviveCovid-19++* to gain deeper understanding about different collaborative efforts taken by people to tackle the *Covid-19* virus. They navigate to the game, join a room and choose their roles among the four roles presented.

The four players are quickly overwhelmed by the waves of virus themed monsters which follow them and infect them upon contact, as depicted in Fig. 1 [A]. As they proceed through the stages, they encounter different varieties of viruses depicting different strains of *Covid-19* as shown in Fig. 1[B]. The four friends use the in-game chat functionality to interact with each other and come up with different strategies to finish the game. Collecting Mask and Sanitization pickups, as indicated by Fig. 1[E], provided the players with temporary protection from the virus as depicted in Fig. 1[F]. Fig. 1[C] depicts player visiting healthcare camps across the map to heal themselves when required, and getting healed. By collecting the primary vaccine parts, depicted in Fig. 1[D], and completing individual goals the players proceed through the multiple stages together and are able to finish the game, all the while carefully following safe guidelines such as social distancing and usage of masks and sanitizers. Upon successful completion of the game the

friends realize that *SurviveCovid-19++* depicts how it would be very difficult to survive without all of the different avatars working together and fulfilling their roles to fight *Covid-19*.

5 EVALUATION AND RESULTS

SurviveCovid-19++ has been developed as a collaborative multiplayer healthcare educational game with an aim to facilitate better understanding about safety measures to be taken against *Covid-19*, importance of roles played by varied professions and about importance and usefulness of taking vaccines. Hence, we evaluate *SurviveCovid-19++* to assess the influence on players' understanding, player experience, and usability. MEEGA+ [18] evaluation model focuses on player experience and usability factors towards evaluating games, and is specifically employed in evaluating educational games. Towards this evaluation, we adapt a mixed methods approach that includes an adapted MEEGA+ questionnaire, accompanied by factors evaluating pedagogical value and influence, presented in [7].

We evaluated *SurviveCovid-19++* with two teams, comprising of four players each. All the players were requested to answer a questionnaire to understand the players' perception about roles played by individuals of varied professions and player perception about need for collaboration to control *Covid-19*. After answering the questionnaire, each team played the game at least once, for fifteen minutes.

After playing the game, the participants were requested to answer two questionnaires, one that was asked before playing the game, to understand perception of the players, and another five-point likert-scale based questionnaire to understand the player experience, usability and pedagogical value dimensions. The questionnaire and results of evaluation are presented here³.

Among the eight volunteers, we observe that four participants did not correctly interpret the role of citizens and law enforcers in controlling the pandemic, based on their answers to the questionnaire asked before they started playing the game. Three participants wrongly interpreted the role of sanitation workers or the need for collaborative efforts towards controlling the pandemic. All the participants described the roles of individuals with different professions correctly to a reasonable extent and also recognized the need for collaborative efforts to control the pandemic after playing the game, indicating a change in their interpretation. This implies that *SurviveCovid-19++* has contributed to better understanding of the players in the context of *Covid-19*.

The mean value was observed to be greater than 3.5 for majority of the quality factors, with low standard deviation (as indicated in Fig. 2), indicating that majority of the players agreed that *SurviveCovid-19++* is useful, interesting and usable.

We calculate the quality score of *SurviveCovid-19++* based on MEEGA+ model by considering the mean values for factors under the dimensions - player experience and usability. Corresponding Cronbach Alpha values presented in MEEGA+ for each dimension are applied to the means of player experience and usability factors, and the resultant value is normalized by multiplying it by 10. This resulted in a quality score of 65.96. According to the MEEGA+ evaluation model, games with quality score greater than 65 are

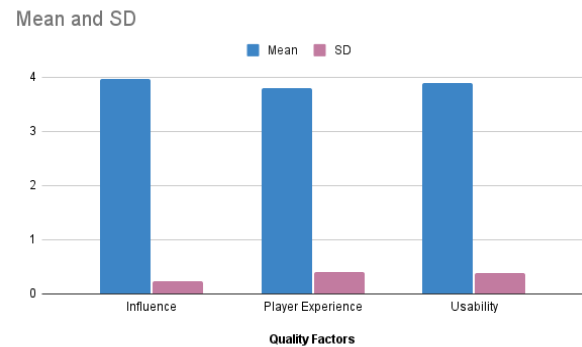


Figure 2: Mean and Standard Deviation Values for the Quality Factors considered for evaluation.

considered to be of excellent quality, and that less than 42.5 are considered to be poor quality. The game has a quality score of 65.96, indicating inclusion of the game in excellent category. This result indicates that such games are challenging for learners, without monotonous activities and clear rules to play the game.

6 DISCUSSION AND LIMITATIONS

The current multiplayer version accommodates only four players per room, analogous to the four avatars being considered in the game and has only four stages. More number of players and stages can be accommodated, with minimal technical modifications in the game.

As we focused on influencing behaviour of the players towards following safety measures for *Covid-19*, and collaborating towards controlling the spread of the pandemic, we performed an in-depth evaluation with eight players and closely observed their behavioural changes when they encounter various scenarios in the game, and assessed their experience with the game. However, an extended evaluation with more number of participants could give us better insights on the impact and usefulness of the game.

While we focused largely on designing the theme of the game, collaborative games, especially in the healthcare domain would have a better impact when designed based on guidelines discussed in the literature. Hence, we see that following certain design guidelines could further positively improve the impact of the game on the players. Furthermore, describing the adaptation of such guidelines in the game could provide insights to game developers and designers on incorporating guidelines into game design.

Moreover, the evaluation presented only specifies adaptation of MEEGA+ model to evaluate the game, but does not explain the factors adapted and the reasons for adapting specific factors. We plan to present these details and the corresponding justifications in future versions of the article.

7 CONCLUSION AND FUTURE WORK

In this paper, we presented *SurviveCovid-19++*, a multiplayer, collaborative desktop game, aimed to facilitate learning about vaccines, safety measures to be followed, roles played by different professions

³<https://github.com/AkhilaSriManasa/SurviveCovid-19ppResults>

(*doctor, citizen, law enforcer and sanitation worker*) and need for collaborative effort to control the spread of *Covid-19*.

SurviveCovid-19++ has been evaluated to assess player experience, usability and its pedagogical value, through a user survey based on an adapted MEEGA+ questionnaire, with two teams of four players each. The results of the evaluation were positive, with observed attitudinal changes and learning outcomes among all the participants, and all the participants willing to recommend *SurviveCovid-19++* to their colleagues.

SurviveCovid-19++ currently supports only four avatars, corresponding to four different professions. More number of avatars could be added in the game, to support wide range of professions in-game, in the future versions. As a part of future work, we also plan to facilitate users to create and customize their own avatars. Also, *SurviveCovid-19++* currently supports only four stages. We plan to increase the number of stages by adapting a procedural approach to generate stages, and increase difficulty in each stage. Moreover, the current version of the game is supported only on desktop, which can be extended to support mobile devices as well.

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